Usability and user experience goals

Introduction

- Part of the process of understanding users is to be clear about the primary objective of developing an interactive product for them
- To help identify the objectives, we suggest classifying them in terms of usability and user experience goals.
- Usability goals are concerned with meeting specific usability criteria, such as efficiency, whereas user experience goals are concerned with explicating the nature of the user experience, for instance, to be aesthetically pleasing.
- https://www.interaction-design.org/literature/topics/usability (Presenter: Alan Dix)

When designing any system: we must clearly define:

- What user should be able to do -> usability
- How user should feel -> UX

Usability Goals

- Usability Goals (What makes a system easy and effective to use)
- focuses on how effectively, efficiently, and satisfactorily a user can achieve tasks using a system.
- **Common Usability Goals:**
- **Effectiveness** – Can users complete tasks accurately?
- **Efficiency** – How quickly and with little effort can tasks be completed?
- **Safety** – Does the system prevent errors and damage?
- **Utility** – Does it provide the right functions? -> how well the system works
- **Learnability** – How easy is it to learn? -> New user
- **Memorability** – Can users remember how to use it later?

Why is safety more important in hospital software that in a game/ music player?

How usability goals can be understood

- Effectiveness: A phone app that helps you book a cab quickly. -> accurately and successfully
- Efficiency: A coffee machine that brews coffee with minimal effort.
- Safety: A child-proof medicine container.
- Utility: A Swiss Army knife (useful in various scenarios).
- Learnability: A user-friendly microwave with simple buttons.
- Memorability: A password manager app with intuitive workflows.
- Satisfaction: An online store with excellent user experience and design.
  - Satisfaction means whether user feel pleased after using the system

User Experience goals (UX)

- User Experience (UX) Goals (How it feels to use)
- goes beyond usability and focuses on the **emotions, perceptions, and satisfaction** users get while interacting with a system.
- **Common UX Goals:**
- Enjoyable ->user likes using it
- Engaging -> it keeps the user interested and involved
- Aesthetically pleasing -> looks attractive and polished
- Motivating -> it encourages user to continue (datacamp.com)
- Satisfying ->user feels good after task completion
- Fun ->in games, children products, and learning applications
- Rewarding -> user feel they gained something meaningful

Key Differences

Aspect	**Usability Goals**	**UX Goals**
Focus	Task performance	User feelings
Nature	Objective (measurable)	Subjective (perception-based)
Example	Time to complete task	Enjoyment level
Question	“Can the user do it?”	“How does the user feel?”

Working Example

- **Example: Food Delivery App**
- **Scenario:** A student orders food using an app.
- **Usability Perspective:**
- Can the student find a restaurant easily? (**Effectiveness**)
- How long does it take to place an order? (**Efficiency**)
- Are there errors in payment or ordering? (**Safety**)
- Is the interface easy to learn? (**Learnability**)

Working Example

- **UX Perspective:**
- Is the app visually appealing?
- Does the user enjoy browsing food options?
- Does the app feel trustworthy and smooth?
- Is the experience satisfying when order is placed?
- A system can be usable but not enjoyable.
  Example: Fast ordering (usable) but boring interface (poor UX).

Quantitative Usability Goals (Measured with numbers)

Goal	Measurement	Example
Efficiency	Time taken	Order placed in < 2 minutes
Effectiveness	Success rate	95% users complete order
Errors	Error rate	< 2 errors per session
Learnability	Time to learn	New user completes task in 1st attempt

Quantitative Usability Goals (Measured with numbers)

- **Example Measurement:**
- 100 users tested
- 90 successfully placed orders

👉 Effectiveness = **90% success rate**

Qualitative Usability Goals (Based on user opinions)

Goal	Measurement Method	Example
Satisfaction	Surveys	“Rate your experience (1–5)”
Ease of use	Interviews	“Was it easy to order?”
Comfort	Observation	User frustration level
Trust	Feedback	“Do you trust this app?”

- These are **subjective and experience-based**.

How to measure(Quantitative)

- **Quantitative Methods:**
- Task completion time
- Error counting
- Click tracking
- Success rate
- **Example:**
- Avg time to order food = 1.5 minutes
- Error rate = 1.2%

👉 Good usability

How to measure(Qualitative)

- User interviews
- Questionnaires (Likert scale)
- Observation
- Think-aloud protocol -> Think-aloud protocol is a usability testing method where users speak their thoughts while using a system.
- **Example (Likert Scale):**
- “The app is easy to use”
  1 (Strongly Disagree) → 5 (Strongly Agree)
- Average score = **4.3** → Positive UX

User says:

- “Where is checkout?”
- “This button is confusing”
- “I think this is cart… oh wait…”

This gives us direct insight into user confusion and expectations

It reveals

- confusion points
- wrong assumptions
- mental model
- usability issues

- **Usability = Can users do the task?**
- **UX = Do users enjoy doing it?**

Best systems achieve both:

- High performance (usable)
- Positive emotions (good UX)

Class Activity

- Case: ATM Machine
- Identify Usability goals
- **Usability Goals:**
- Withdraw cash in < 1 minute
- Minimal errors
- Easy navigation

- Identify UX goals
- **UX Goals:**
- Feeling of security
- Clear instructions
- Stress-free interaction

Observation

- **Observation:**
- If ATM is fast but confusing → good usability, poor UX
- If ATM is attractive but slow → good UX, poor usability

Score the product on each of these factors (out of 5)

Usability Goal	Criteria		Product A Score	Product B Score
Effectiveness	Does it accomplish its goal?	How well does it solve the problem?		
Efficiency	Does it save time?	Is it resource-efficient?		
Safety	Are there risks or hazards?	Does it prevent errors?		
Utility	How useful is it in different contexts?	Is it versatile?		
Learnability	How easy is it to learn?	Can new users grasp it quickly?		
Memorability	Can users easily remember how to use it after a break?	Is the design memorable?		
Satisfaction	How enjoyable is it to use?	Does it leave users satisfied?		

Quantitative Usability Goals - Example 1

Usability Goal	Metric	Value (Quantitative)
Effectiveness	Task completion rate	92%
Efficiency	Average time on task	3 minutes
Safety	Error rate	1 error per 50 uses
Utility	Feature usage rate	75% of features used
Learnability	Time to proficiency	5 minutes per user
Memorability	Task relearning time after 1 month	1 minute
Satisfaction	SUS score (system usability scale)	87/100

Quantitative Usability Goals - Example 2

Usability Goals

Goal #: 1
Task: Log on with security

Ease-of-Learning Goals

Priority	Measure	Goal
	Novice Time	
1	Novice Trials	3
	Novice Errors	

Operational Definitions
Expert: Third trial
Novice: First two trials
Learn: Error-free performance
Satisfaction: 1=very unsatisfactory
4=neutral, 7=very satisfactory

Ease-of-Use Goals

Priority	Measure	Goal
1	Expert Time	$\leq$ 7 sec.
1	Expert Errors	0

Priority Definitions
1 = Required for release
2 = Important if not too ##
3 = Desirable if easy

Satisfaction Goals

Priority	Measure	Goal
3	Expert	6
3	Novice	5

User Experience Goals

Desirable aspects		
Satisfying	Helpful	Fun
Enjoyable	Motivating	Provocative
Engaging	Challenging	Surprising
Pleasurable	Enhancing sociability	Rewarding
Exciting	Supporting creativity	Emotionally fulfilling
Entertaining	Cognitively stimulating	Experiencing flow
Undesirable aspects		
Boring	Unpleasant	
Frustrating	Patronizing	
Making one feel guilty	Making one feel stupid	
Annoying	Cutesy	
Childish	Gimmicky	

Table 1.1 Desirable and undesirable aspects of the user experience

User Experience Goals (Cont'd)

- The two lists are not meant to be exhaustive.
- There are likely to be more—both desirable and undesirable—as new products surface.
- The reason for there being more of the former is that a primary goal of interaction design is to create positive experiences. There are many ways of achieving this.
- Not all usability and user experience goals will be relevant to the design and evaluation of an interactive product being developed. Some combinations will also be incompatible